

To whom it may concern.

São Carlos (Brazil), 7th of April, 2026

Subject: Recommendation letter for MSc. António José Freitas Machado Dórdio Gomes

I am pleased to write this letter of recommendation for António José Freitas Machado Dórdio Gomes, whom I have known for approximately the past two and a half years as a student in the Master's programme in Mechanical Engineering at the Faculty of Engineering of the University of Porto (FEUP). During this period, I had the opportunity to interact with Mr. Gomes both as his professor in courses related to aeronautical structural design and analysis, and as the supervisor of his Master's thesis.

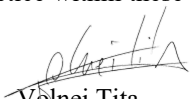
Over this time, he consistently demonstrated strong analytical skills, technical maturity, and a rigorous engineering mindset. His work has been marked by a high level of commitment and intellectual curiosity, as well as the ability to approach complex engineering problems in a structured and methodical way. During the courses Aeronautical Structural Design and Analysis, Mr. Gomes developed extensive work on the structural analysis of an aircraft wing-box, addressing both static and fatigue behaviour. These projects involved the development of detailed finite element models using ABAQUS, the implementation of aerodynamic load cases, and the assessment of the structural response according to certification-oriented design philosophies such as infinite life, safe-life, and damage tolerance approaches. The work required the integration of several aspects of structural mechanics, including composite structural analysis, fatigue assessment, and fracture mechanics, together with careful modelling and interpretation of the numerical results. Throughout these projects, he showed a clear ability to translate theoretical concepts into practical computational models, while maintaining a critical engineering perspective when analyzing the results.

His Master's thesis, entitled "Low-Velocity Impact Damage Assessment and Numerical Modelling of Glass/PPS Laminated Composites Using Commercial Finite Element Tools", further confirmed his ability to carry out research in the field of aerospace composite structures. In this work, Mr. Gomes developed a numerical modelling framework using Abaqus/Explicit to investigate the impact response and damage evolution in thermoplastic composite laminates. The study included the implementation of intralaminar damage modelling using Hashin criteria, cohesive zone modelling for interlaminar delamination, and a bi-phasic modelling strategy, together with a dedicated investigation of the influence of matrix plasticity on the overall impact response. The numerical results were systematically compared with experimental impact data and non-destructive testing measurements, allowing a quantitative assessment of the damage mechanisms involved. One aspect I would particularly highlight in his work is the ability to conduct complex technical studies with a high degree of autonomy. During part of his thesis work carried out abroad he benefited from close technical guidance, but after returning to Portugal he continued the development of the numerical models and analyses largely independently. This transition clearly demonstrated his initiative, perseverance, and ability to critically evaluate modelling strategies and simulation results while maintaining a rigorous and methodical approach to engineering problems.

Beyond his technical capabilities, Mr. Gomes stands out for his initiative, reliability, and ability to communicate technical ideas clearly and effectively. He is a motivated and disciplined engineer who is capable of managing complex technical tasks and delivering well-structured results.

Based on my experience supervising his work over the past years, I am confident that he possesses the technical preparation, intellectual curiosity, and professional attitude required to succeed in any engineering environment, particularly in fields related to aeronautical structures, structural simulation, and composite materials. For these reasons, I recommend him without reservation for opportunities in research, development, or engineering practice within these areas. Please feel free to contact me should you require any further information.

Sincerely yours,



Volnei Tita
(Eng, MSc, Dr.Eng)

Associate Professor
University of São Paulo - voltita@sc.usp.br

Affiliated Professor
University of Porto – UP - voltita@fe.up.pt

To whom it may concern,

Milano, 15th May 2026

I am pleased to recommend António José Freitas Machado Dordio Gomes, who carried out their thesis work under my supervision in the Aerospace Science and Technology Department of Politecnico di Milano, Italy, within the Erasmus program.

The thesis focused on finite element modeling (FEM) of impact phenomena in glass/PPS composite materials, with particular emphasis on the implementation and calibration of explicit numerical models. Throughout the project, the student demonstrated a high level of independence, systematically addressing complex numerical modeling challenges.

They showed strong capability in handling and comparing different constitutive laws, as well as a solid understanding of explicit approaches for highly dynamic analyses. The work was conducted with methodological rigor, critical interpretation of results, and careful attention to numerical validation aspects.

Overall, António has demonstrated sound technical skills and a clear aptitude for research, making them well-suited for industrial or academic environments in the field of solid mechanics and computational modeling.

Sincerely,

Antonio Maria Caporale, PhD
Assistant Professor



Subject: Letter of recommendation – António Gomes

During the first half of 2023, António José Freitas Machado Dórdio Gomes, a student of Bachelor's Degree in Mechanical Engineering at the University of Porto, took part of a curricular internship at the Research, Development and Certification section of WEG Portugal, an industrial electric motors manufacturer, part of the multinational group WEG.

This internship aimed at the study and specification of a thermal conductivity effective method of permanently bond fins to steel frames of electric motors, and was directly oversighted by me. The project had two components: a simulation one, in which António had to use the Ansys software, and a practical one, which required the use of a test setup to compare results. In that time, António demonstrated a remarkable dedication to the project, always wanting to go beyond than expected, and good analytical and scientific skills, often offering inputs that increased the added value of the project. In interpersonal relationships, António has shown autonomy and good communications skills, and was quickly integrated not only in the project team, but also within the company.

For the reasons mentioned above, it is my opinion that António Gomes has the necessary skills, knowledge and qualities to excel in any junior mechanical engineer position. I am free to answer any questions regarding António's performance at jorgefs@weg.net.

Maia, 7 July 2023

A handwritten signature in blue ink that reads 'Jorge Fernando de Sousa'. The signature is fluid and cursive, with a horizontal line drawn underneath it.

Jorge Fernando de Sousa
Mechanical Engineer, Research and Development at WEG

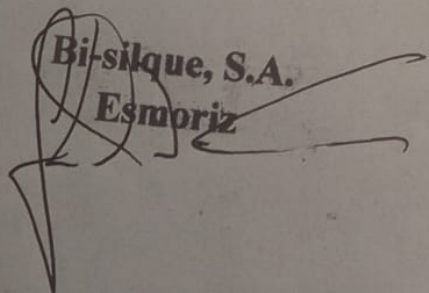
Carta de recomendação

O Sr. António José Freitas Machado Dordio Gomes, efetuou um estágio na nossa empresa no setor de manutenção com início em 01-08-2022 e termo a 02-09-2022.

Durante o estágio, desempenhou sempre as suas tarefas de forma eficiente, demonstrando uma elevada competência, demonstrou sempre vontade de aprender, nunca se coibindo de adquirir novos conhecimentos, evidenciando também um excelente espírito de equipa.

Em suma, consideramos que o Sr. António Gomes demonstrou excelentes atributos como honestidade, competência e profissionalismo.

Esmoriz, 02/09/2022.



Bi-silque, S.A.
Esmoriz

Rua Cais da Estação,
3885-528 Esmoriz
P.O. BOX 43, 3886-909 Esmoriz
Portugal

Sociedade Anónima, registo na c.r.c. de Ovar
e NIF 500924902.
Capital Social: 2.500.000 Euro int. realizado

Bi-silque S.A. is part of Bi-silque SGPS

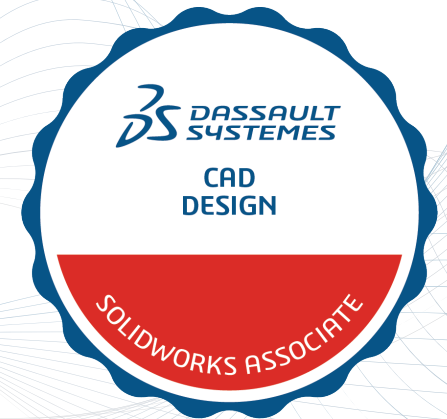
Export Department
Telef. +351 256 750130
Fax +351 256 751141
info@bisilque.com

Departamento Nacional
Telef. +351 256 750120
Fax +351 256 750129
geral@bisilque.com



CERTIFICATE

Dassault Systèmes confers upon
ANTÓNIO GOMES
the certificate for
SOLIDWORKS CAD Design Associate



August 20 2023

Academic exam at Faculty of Engineering, University of Porto

A handwritten signature in black ink, appearing to read 'Manish Kumar'.

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President



C-2JAUPXEG85

Antonio Gomes

is certified to have completed the Composites Design Workshop XXVII
and seen ways to make lighter structures with double-double.

July 22 – 26, 2024



Stephen W. Tsai
Professor Emeritus